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AI-Powered
Personal Finance Intelligence System

Team Name : House of Coders

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Abstract

Problem

Managing personal finances is becoming increasingly complex due to multiple investment options such as stocks, mutual funds, fixed deposits, and loans. However, most financial tools only provide basic tracking and lack intelligent insights.

Our Approach

This project proposes an AI-Powered Personal Finance Intelligence System that allows users to securely store their financial data and interact with an AI assistant through natural language. The system analyzes structured financial data, integrates market trends, and generates personalized financial insights.

Value

The platform also supports financial scenario simulation, enabling users to explore how different investment strategies may impact their long-term financial growth.

Introduction

"Compound interest is the eighth wonder of the world. He who understands it, earns it... he who doesn't, pays it."

-Albert Einstein

Existing finance apps mainly show past spending through basic dashboards but lack intelligent guidance on what users should do next.

Today, users invest in multiple financial instruments such as:

- Stocks
- Mutual funds
- Fixed deposits
- Retirement funds
- Loans and EMIs

Our solution introduces an AI-powered financial assistant that analyzes a user's financial profile, integrates market data, and provides personalized recommendations, predictions, and scenario simulations.

This helps users understand their portfolio performance, financial health, and long-term financial strategies in a simple conversational interface.

Problem Statement

An AI-Powered Personal Finance Intelligence System that integrates financial data, real-time market insights, and AI-driven analysis to provide personalized financial insights, scenario simulations, and smarter investment recommendations.

Target Users

- Young professionals managing early investments
- Retail investors participating in stock and mutual fund markets
- Individuals with diversified financial assets
- Users planning long-term goals such as retirement or wealth accumulation

Limitations of Existing Solutions

- Fragmented financial data across multiple platforms
- Lack of AI-driven financial reasoning and insights
- No support for scenario simulation and predictive financial planning

Proposed Solution

To address these challenges, we propose an AI-Powered Personal Finance Intelligence System that integrates structured financial data, real-time market insights, and AI-driven reasoning to provide personalized financial analysis, scenario simulations, and strategic investment recommendations.

Technical Details

Frontend

NEXT.JS

Tailwind CSS

shadcn/ui

Creates a responsive and user-friendly interface for smooth interaction.

Backend

node JS

FastAPI

Handles API processing and system logic efficiently.

AI Layer

Gemini

LangChain

{REST}

Analyzes data to generate intelligent insights and recommendations.

Database

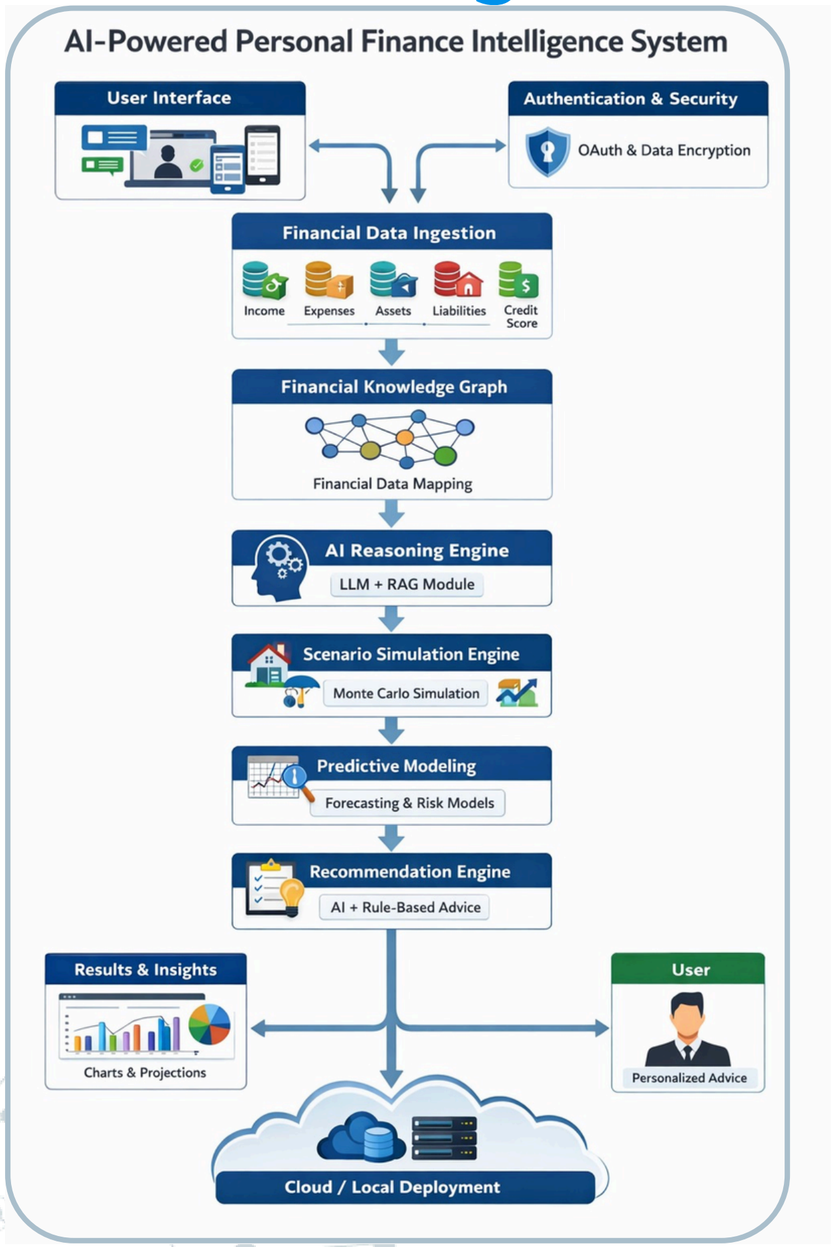
PostgreSQL

redis

mongoDB

Manages secure storage and quick data retrieval.

System Design



1. Securely collects financial data such as income, expenses, assets, liabilities, and credit scores.
2. Organizes information using a Financial Knowledge Graph for structured relationships.
3. An AI Reasoning Engine (LLM + RAG) analyzes user financial profiles.
4. Monte Carlo simulations and predictive models estimate future financial scenarios and risks.
5. The system generates personalized financial recommendations and visual insights.
6. Deployed on cloud or local infrastructure for scalable and secure access.

Business Prospect

The demand for AI-driven financial advisory platforms is growing rapidly as more individuals participate in stock markets and digital investments.

Market Relevance

- Increasing retail investors in stock markets
- Growing demand for digital financial advisory
- Rise of AI-driven decision support systems

Potential Users

- Individual investors
- Financial startups
- Investment advisory firms
- Banks and fintech companies

Unique Selling Point

- Unlike traditional financial apps, this system provides:
- AI-based financial reasoning
- Real-time portfolio analysis
- Financial scenario simulation
- Personalized investment strategies

Monetization Strategies

- Freemium Model
- Free portfolio tracking
- Premium AI financial insights
- Subscription Model
- Monthly financial planning reports
- Advanced analytics
- B2B Integration
- Integration with fintech platforms
- Financial advisory tools for banks

Conclusion

Impact & Key Takeaways

Proactive vs. Reactive: Users shift from looking at past mistakes to simulating future successes. Accuracy over Hallucination: By offloading math to a Python sandbox, we ensure 100% financial reliability.
Privacy as a Priority: Our PII masking architecture ensures that "Personal Finance" stays truly personal.

Future Scope

Live Integration: Connecting via Open Banking APIs for real-time automated updates. Tax **Optimization**: Automating tax-saving strategies based on regional laws. Multi-Agent **Ecosystem**: Integrating with AI agents from real estate and stock brokers for end-to-end execution.

Benefits

Users: Access to "Virtual CFO" level advice without the high cost of human advisors.
Industry: Provides a blueprint for "Zero-Knowledge" financial AI that banks and FinTechs can safely adopt.
Society: Direct impact on financial literacy and wealth inequality by making complex optimization accessible to everyone.

References

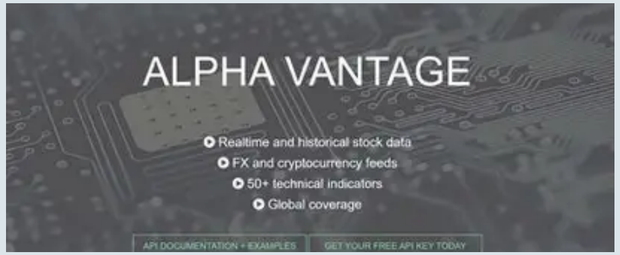
Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks

Deep Portfolio Theory

Yahoo Finance

FAISS

Gemini API documentation



Alpha Vantage